Praktikum Elektronika Analog

Mendesain Penguat Sinyal Kecil (AC) Rangkaian Fixed Bias Common Emitter BJT

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Tujuan Praktikum

1. Memahami karakteristik BJT dan area kerjanya
2. Memahami parameter yang dibutuhkan pada datasheet untuk mendesain rangkaian BJT
3. Mampu mendesain rangkaian sederhana fixed bias pada konfigurasi common emitter BJT untuk beroperasi di daerah aktif
4. Mampu merangkai rangkaian sederhana fixed bias pada konfigurasi common emitter BJT di protoboard
5. Mampu menghitung penguatan dan analisis sinyal kecil AC pada rangkaian BJT

Peralatan yang dibutuhkan

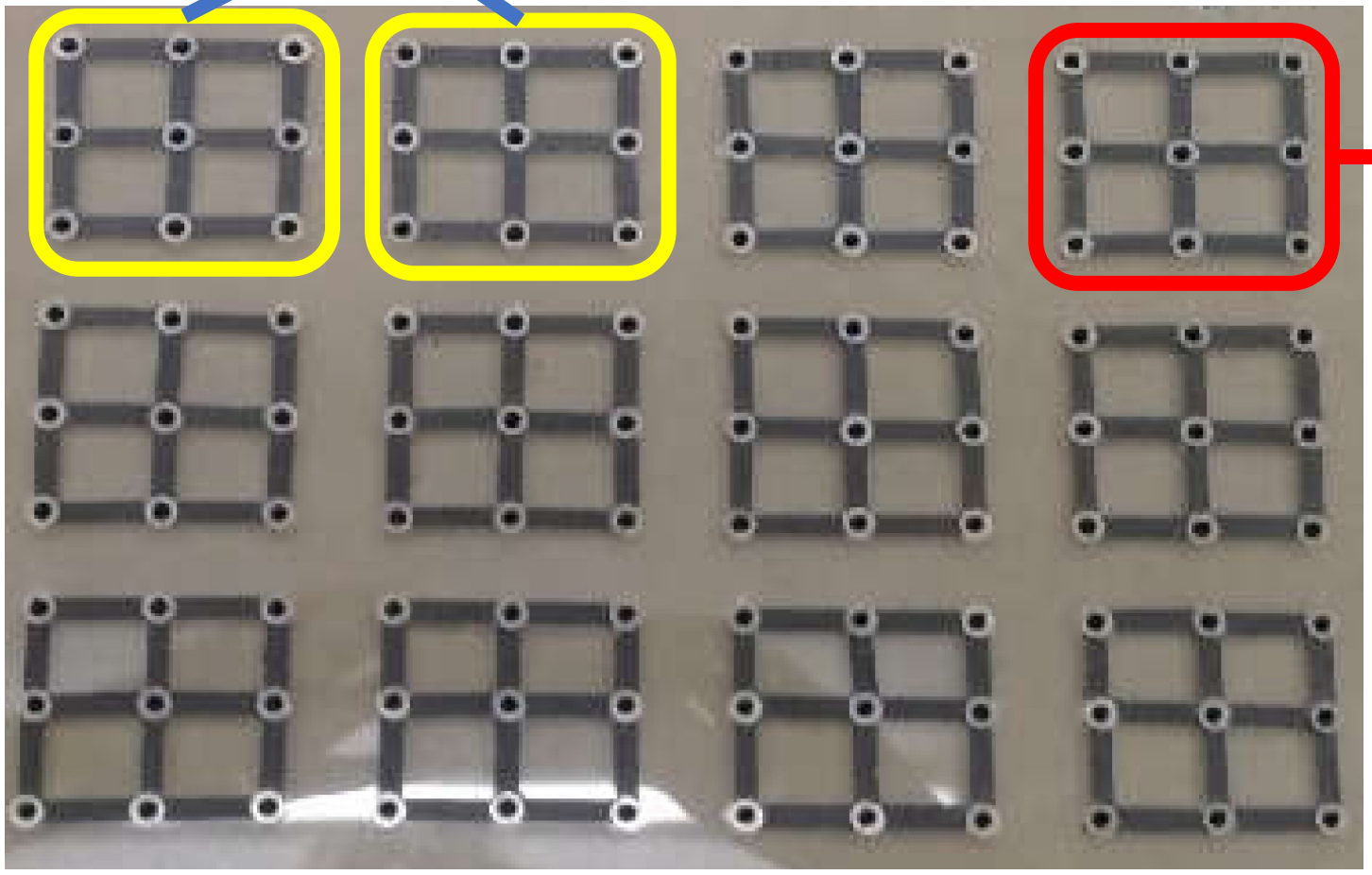
1. DC Power Supply (1)
2. Function Generator (1)
3. Oscilloscope (1)
4. BC107/BC108 NPN Transistor atau padanannya TO-92 (1)
5. Resistor (2 buah sesuaikan dengan desain perhitungan)
6. Capasitor = 10µF (2)
7. Multimeter (2)
8. Kabel Jumper (secukupnya)
9. Konektor Jumper (secukupnya)
10. Kertas HVS dan millimeter block (masing-masing 1 lembar/kelompok)

Persiapan praktikum yang harus dilakukan

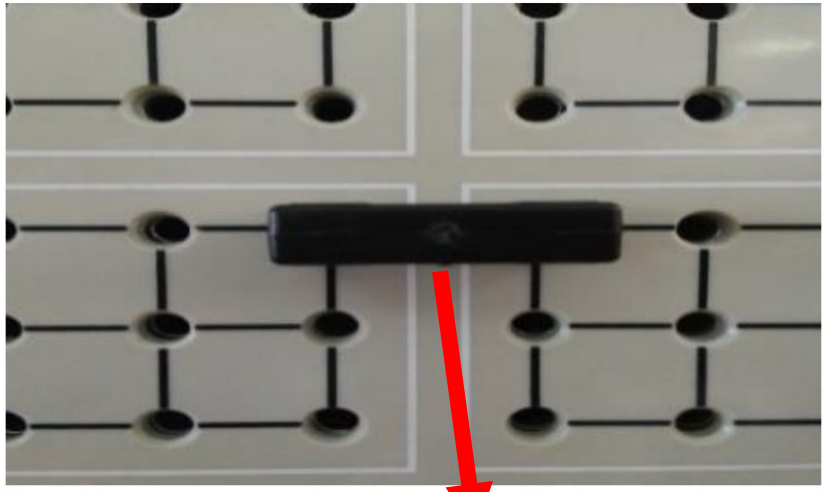
1. Melakukan pengecekan kabel
2. Melakukan pengecekan transistor
3. Merangkai BJT fixed bias configuration dengan benar sesuai dengan desain yang telah dibuat
4. Memperhatikan dan menerapkan K3

Pengenalan Protoboard

Tidak saling terhubung

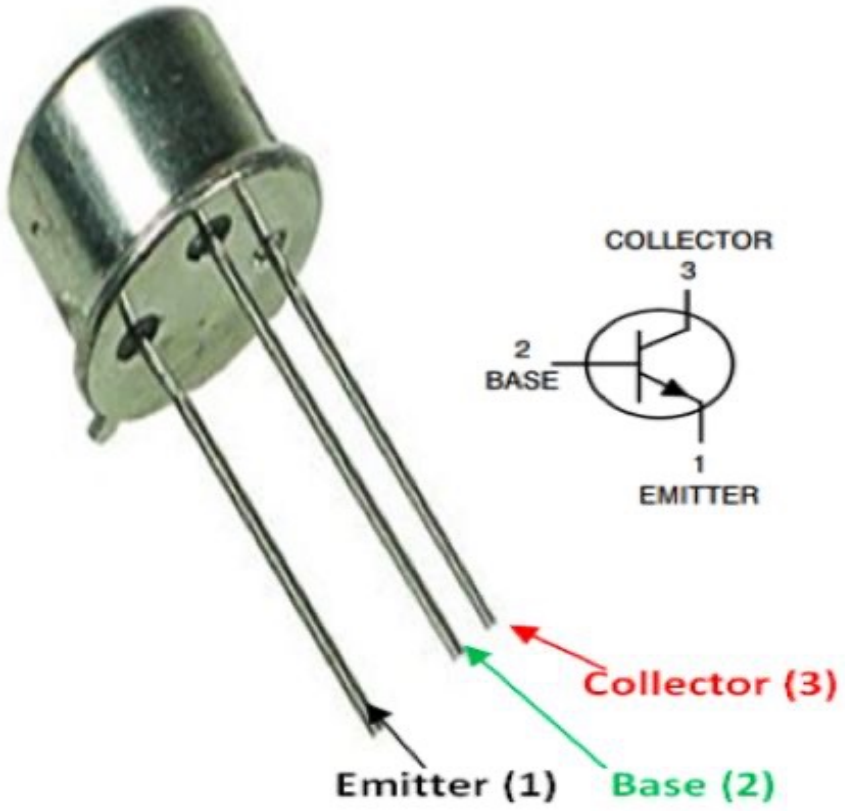


Semua titik saling terhubung



Contoh menghubungkan
2 kelompok matrix dengan jumper/konektor

Cara mengecek BJT dengan multimeter



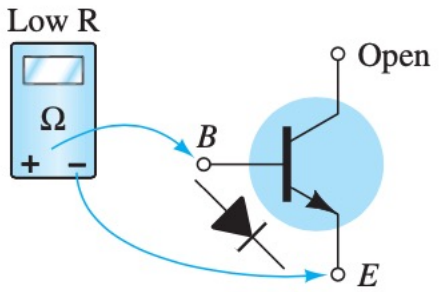


FIG. 3.27
Checking the forward-biased base-to-emitter junction of an npn transistor.

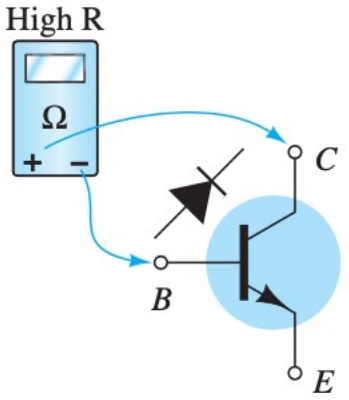


FIG. 3.28
Checking the reverse-biased base-to-collector junction of an npn transistor.

Datasheets (1/3)

BC107/BC108 Series

Low Power Bipolar Transistors

multicomp

Absolute Maximum Ratings

Description	Symbol	BC107	BC108	Unit
Collector-Emitter Voltage	V_{CEO}	45	25	V
Collector-Base Voltage	V_{CBO}	50	30	
Emitter-Base Voltage	V_{EBO}	6.0	5.0	
Collector Current Continuous	I_C	0.2		A
Power Dissipation at T_a = 25°C Derate above 25°C	P_D	0.6 2.28		W mW/°C
Power Dissipation at T_C = 25°C Derate above 25°C		1.0 6.67		
Operating and Storage Junction Temperature Range	T_J, T_{stg}	-65 to +200		°C
Thermal Resistance				
Junction to Case	$R_{th\ (j-c)}$	175		°C/W

Datasheets (2/3)

Electrical Characteristics (T_a = 25°C unless otherwise specified)

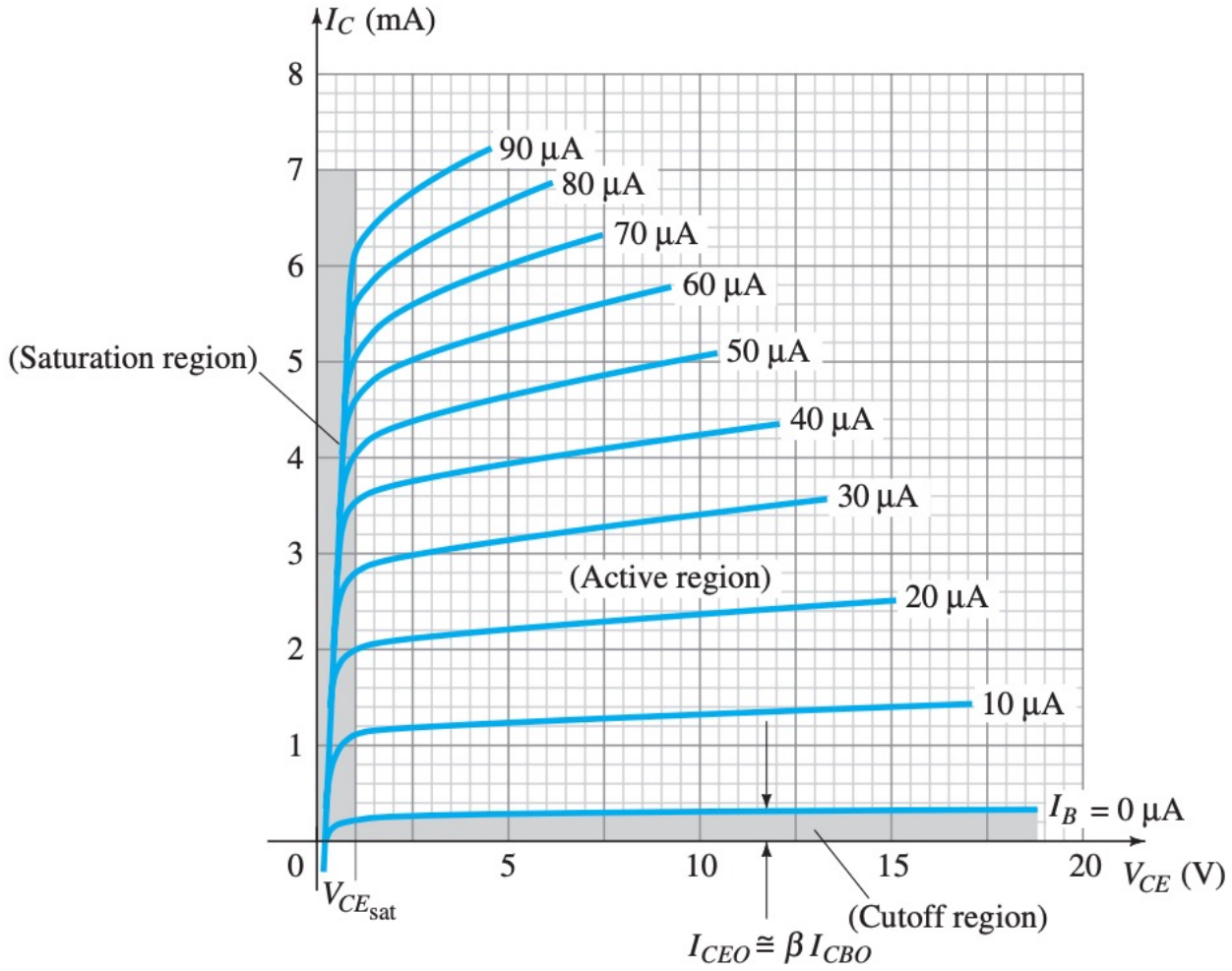
Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector-Emitter Voltage	V_{CEO}	I_C = 2mA, I_B = 0 BC107 BC108	45 25	-	V
Emitter-Base Voltage	V_{EBO}	I_E = 10μA, I_C = 0 BC107 BC108	6.0 5.0	-	V
Collector-Cut off Current	I_{CBO}	V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108 T_{amb} = 125°C V_{CB} = 45V, I_E = 0 BC107 V_{CB} = 25V, I_E = 0 BC108	-	15 15 4.0 4.0	nA μA
DC Current	h_{FE}	I_C = 10μA, V_{CE} = 5V **B Group** **C Group** I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	40 100 110 110 110 200 420	- - 450 800 220 450 800	-
Base Emitter Saturation Voltage	$V_{BE\ (Sat)}$	I_C = 10mA, I_B = 0.5mA I_C = 100mA, I_B = 5mA	-	0.83 1.05	V
Collector Emitter Saturation Voltage	$V_{CE\ (Sat)}$		-	0.25 0.60	V
Base Emitter On Voltage	$V_{BE\ (on)}$	I_C = 2mA, V_{CE} = 5V I_C = 10mA, V_{CE} = 5V	0.55 -	0.70 0.77	V

Datasheets (3/3)

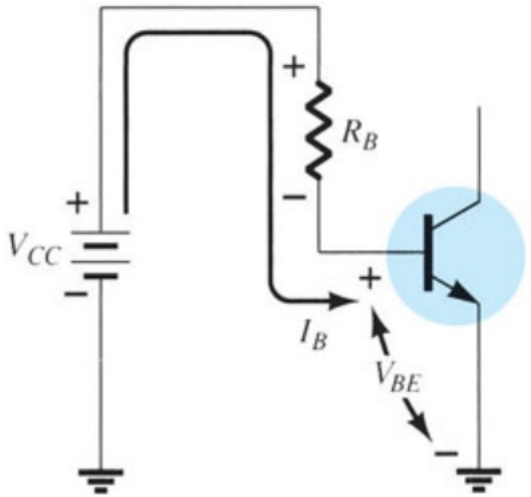
Electrical Characteristics (T_a = 25°C unless otherwise specified)

Description	Symbol	Test Condition	Minimum	Maximum	Unit
Collector Knee Voltage	$V_{CE\ (K)}$	I_C = 10mA, I_B = The value for which I_C = 11mA at V_{CE} = 1V	-	0.60	V
Transition Frequency	f_t	V_{CE} = 5V, I_C = 10mA, f = 100MHz	150	-	MHz
Noise Figure	NF	V_{CE} = 5V, I_C = 0.2mA R_g = 2kΩ F = 1kHz,B = 200Hz		10	dB
Output Capacitance	C_{obo}	V_{CB} = 10V, f = 1MHz	-	4.5	pF
Small Signal Current Gain	h_{fe}	ALL f = 1kHz I_C = 2mA, V_{CE} = 5V **BC 107** **BC 108** **A Group** **B Group** **C Group**	 125 125 125 240 450	 500 900 260 500 900	-
Input Impedance	h_{ie}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	 1.6 3.2 6.0	 4.5 8.5 15	 KΩ KΩ
Output Admittance	h_{oe}	I_C = 2mA, V_{CE} = 5V **A Group** **B Group** **C Group**	-	 30 60 110	umhos

Kurva Karakteristik Output Common Emitter



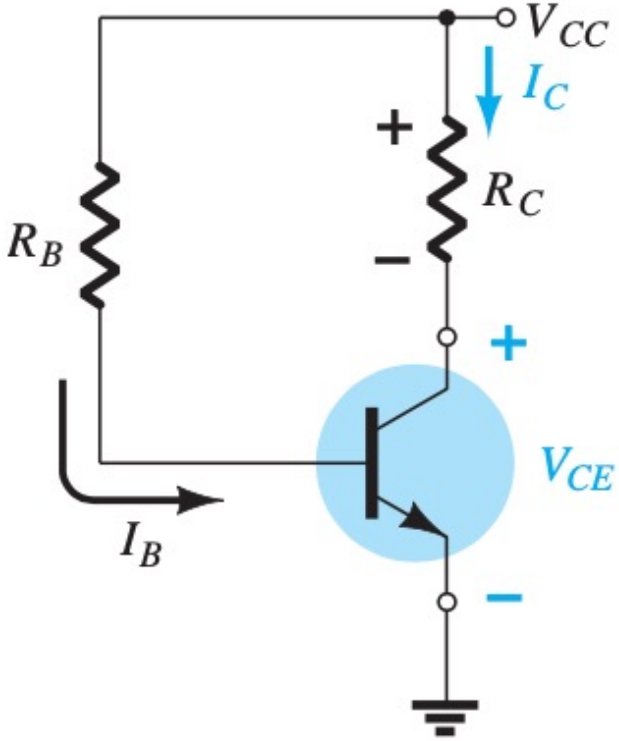
Analisis DC Fixed Bias

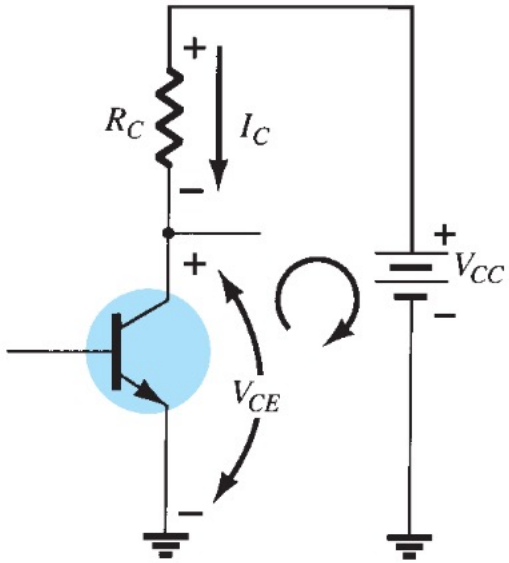


$$+V_{CC} - I_B R_B - V_{BE} = 0$$

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \beta I_B$$





$$V_{CE} + I_C R_C - V_{CC} = 0$$

$$V_{CE} = V_{CC} - I_C R_C$$

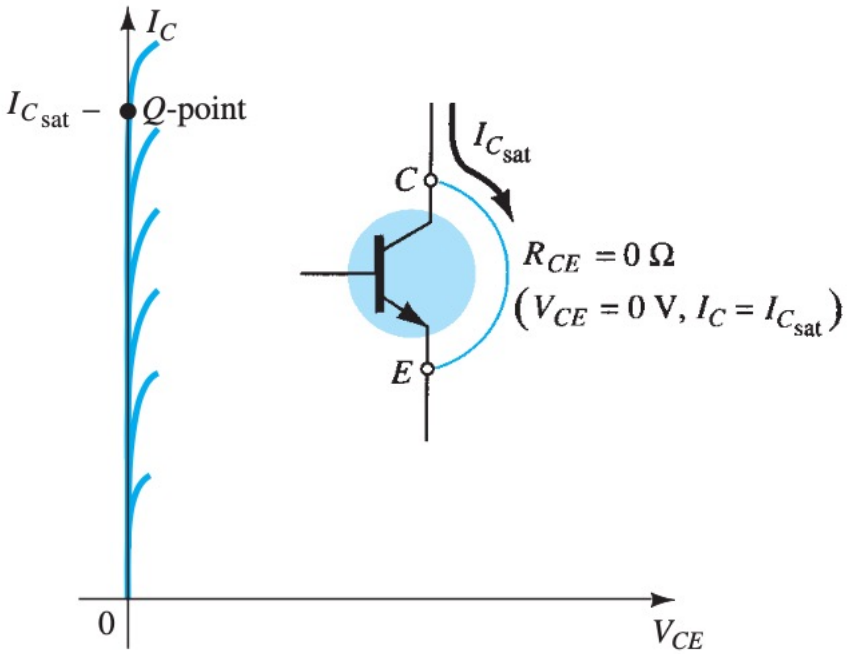
Menentukan Arus Collector Maksimum

Transistor Saturation

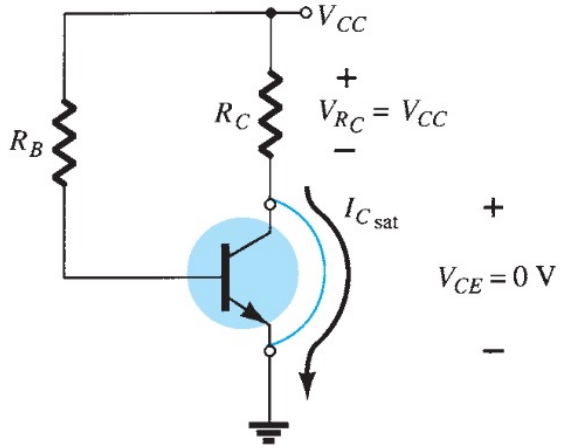
The term *saturation* is applied to any system where levels have reached their maximum values.

$$R_{CE} = \frac{V_{CE}}{I_C} = \frac{0\ \text{V}}{I_{C_{sat}}} = 0\ \Omega$$

$$I_{C_{sat}} = \frac{V_{CC}}{R_C}$$



Approximate



PENTING!!!!

Setelah I_{Csat} diketahui, kita mendapatkan gambaran arus collector maksimum untuk desain yang digunakan. Jika ingin mendapatkan amplifikasi linier, maka titik kerja arus collector harus berada di bawah level I_{Csat}

Garis Beban dan Penentuan Titik Kerja

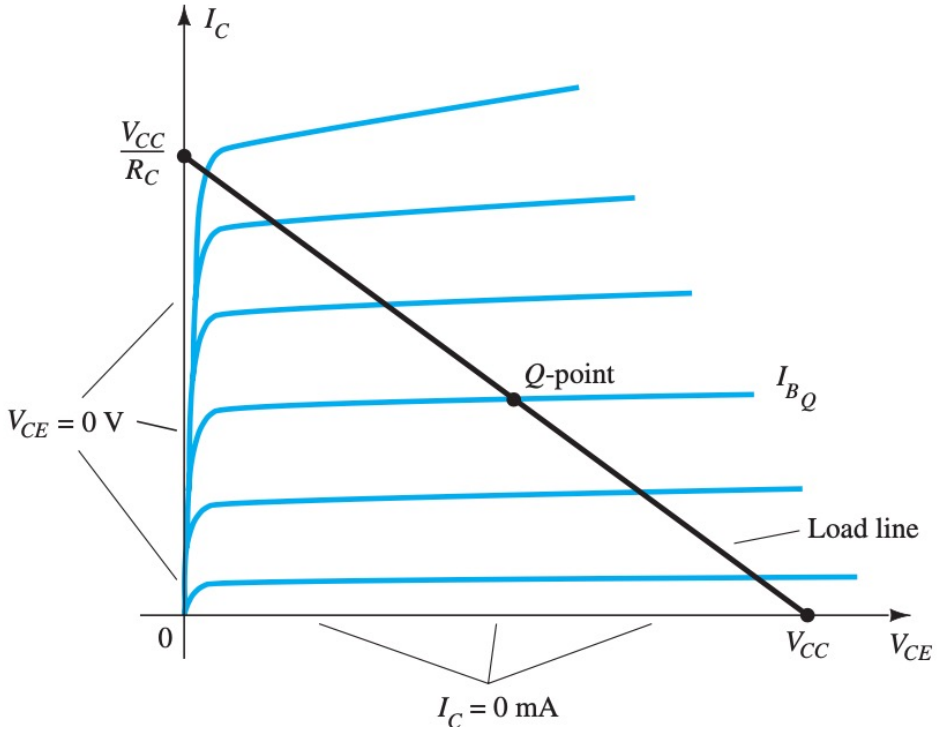
$$V_{CE} = V_{CC} - I_C R_C$$

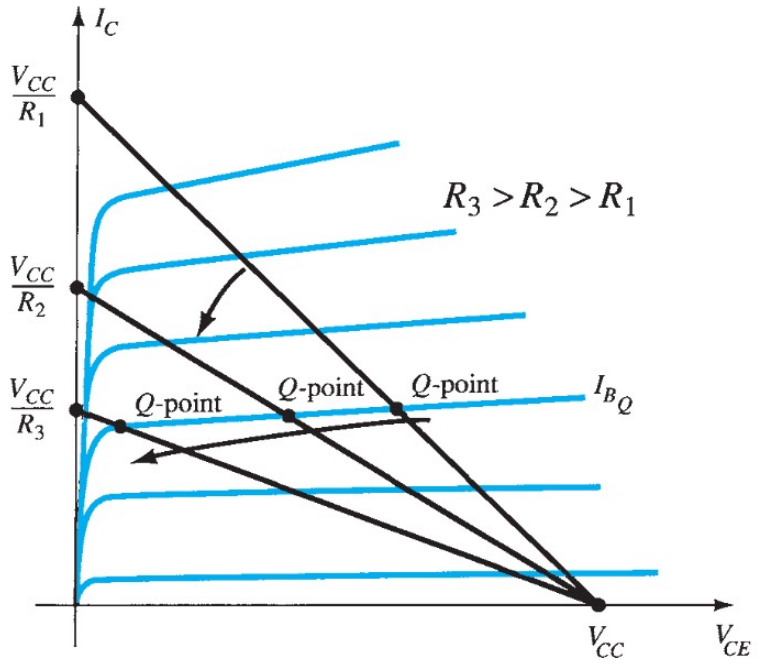
$$V_{CE} = V_{CC} - (0)R_C$$

$$V_{CE} = V_{CC}|_{I_C=0\text{ mA}}$$

$$0 = V_{CC} - I_C R_C$$

$$I_C = \frac{V_{CC}}{R_C}\bigg|_{V_{CE}=0\text{ V}}$$

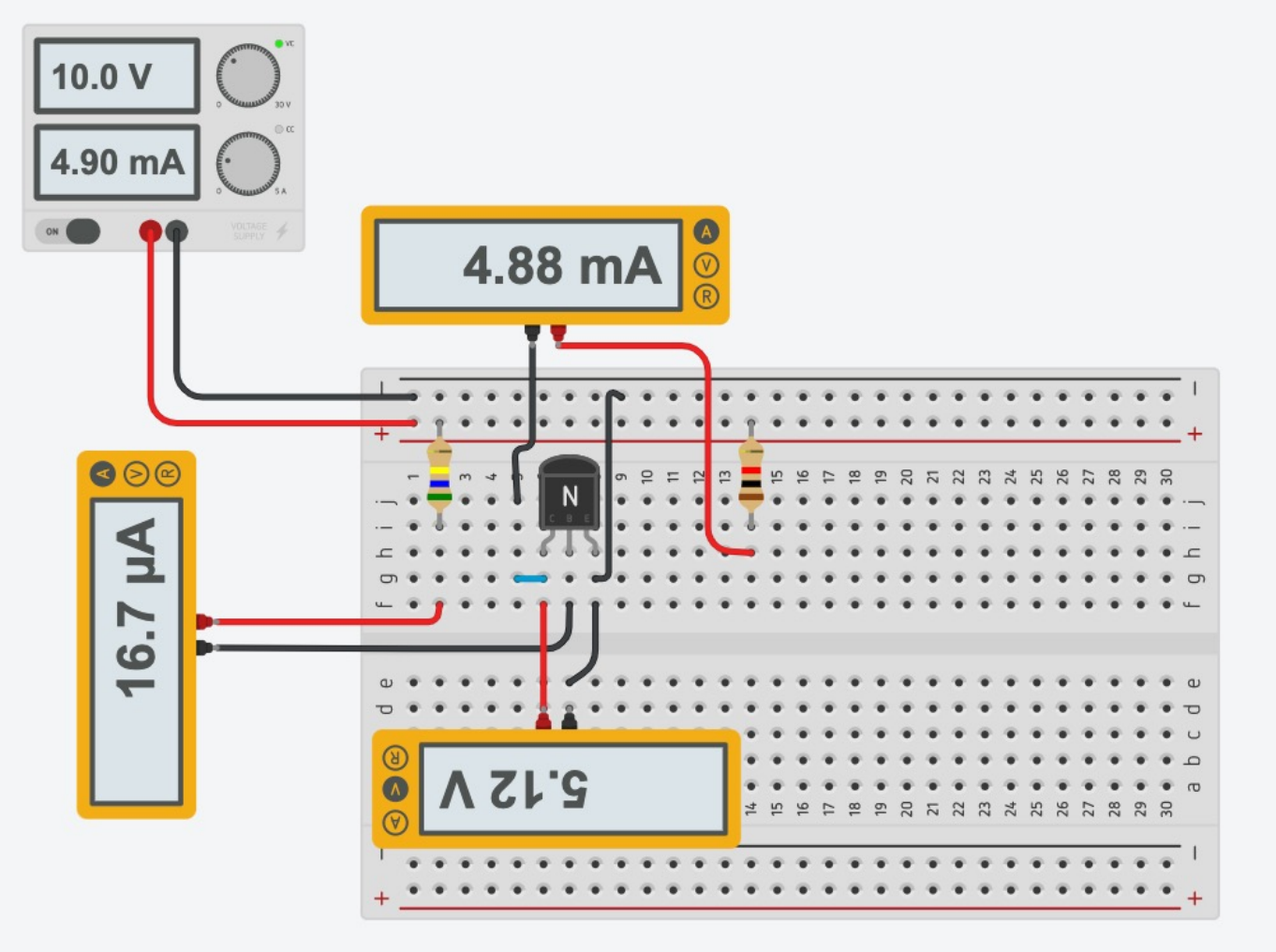




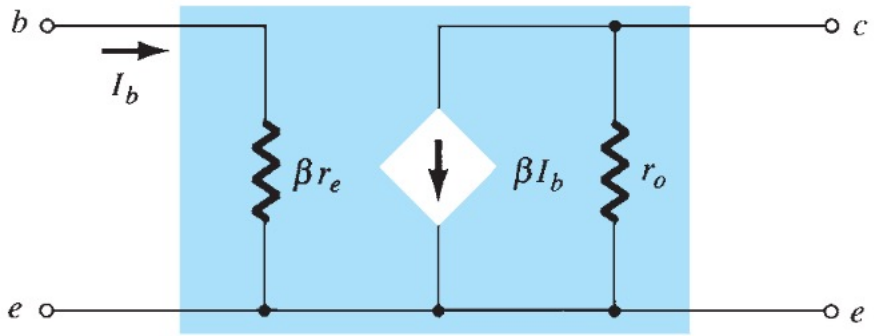
Pengaruh nilai R terhadap Garis Beban

Diagram wiring pengukuran (simulasi) untuk analisis DC

V_{CC} = 10 V
I_{Csat} = 10mA
R_C=1K
I_{CQ} = 5mA
β =292
I_B = 16.7uA
R_B = 556886 = 560K Ohm



Rangkaian Ekivalen BJT Common Emitter



$$r_e = 26\,\text{mV}/I_E$$

$$Z_i = \frac{V_i}{I_b} = \frac{V_{be}}{I_b}$$

$$V_{be} = I_e r_e = (I_c + I_b) r_e = (\beta I_b + I_b) r_e$$
$$= (\beta + 1) I_b r_e$$

$$Z_i = \frac{V_{be}}{I_b} = \frac{(\beta + 1) I_b r_e}{I_b}$$

$$\boxed{Z_i = (\beta + 1) r_e \cong \beta r_e}$$

Resistor Color Code

Quick Guide

1st Band (F)
2nd Band (S)
Multiplier (M)

Color	Value
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Gray	8
White	9

Band 3	Divide by
Gold	10
Silver	100

4- Band Code



100 Ω ± 5%

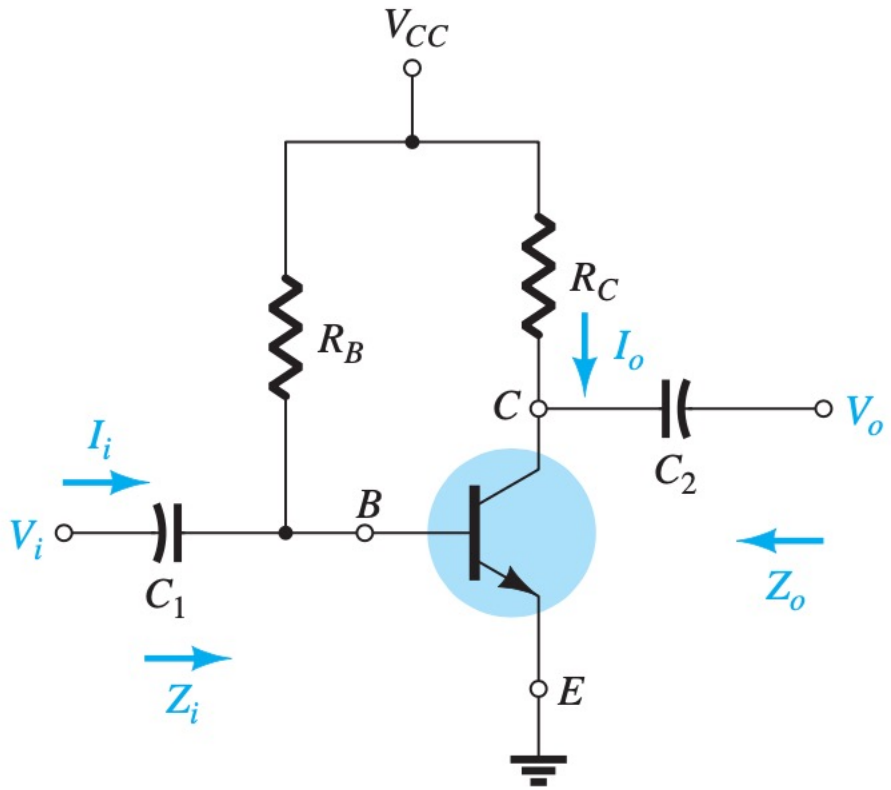
OR

Tolerance	
Brown	± 1 %
Red	± 2 %
Gold	± 5 %
Silver	± 10 %
None	± 20 %

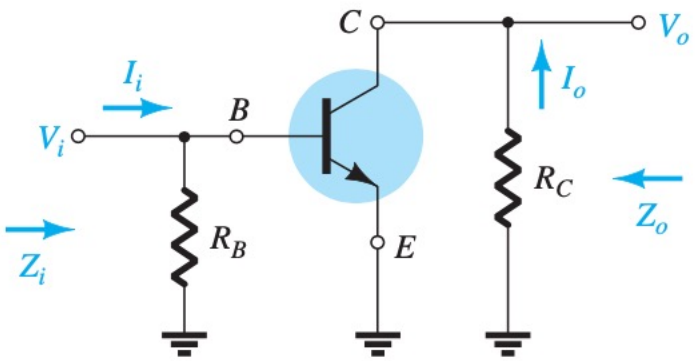
F S M								
.22 ohm								
.27 ohm								
.33 ohm								
.39 ohm								
.47 ohm								
.56 ohm								
.68 ohm								
.82 ohm								
1.0 ohm	10 ohm	100 ohm	1.0 k	10 k	100 k	1 M	10 M	
1.1 ohm	11 ohm	110 ohm	1.1 k	11 k	110 k	1.1 M		
1.2 ohm	12 ohm	120 ohm	1.2 k	12 k	120 k	1.2 M		
1.3 ohm	13 ohm	130 ohm	1.3 k	13 k	130 k	1.3 M		
1.5 ohm	15 ohm	150 ohm	1.5 k	15 k	150 k	1.5 M		
1.6 ohm	16 ohm	160 ohm	1.6 k	16 k	160 k	1.6 M		
1.8 ohm	18 ohm	180 ohm	1.8 k	18 k	180 k	1.8 M		
2.0 ohm	20 ohm	200 ohm	2.0 k	20 k	200 k	2.0 M		
2.2 ohm	22 ohm	220 ohm	2.2 k	22 k	220 k	2.2 M		
2.4 ohm	24 ohm	240 ohm	2.4 k	24 k	240 k	2.4 M		Resistor Values
2.7 ohm	27 ohm	270 ohm	2.7 k	27 k	270 k	2.7 M		
3.0 ohm	30 ohm	300 ohm	3.0 k	30 k	300 k	3.0 M		
3.3 ohm	33 ohm	330 ohm	3.3 k	33 k	330 k	3.3 M		
3.6 ohm	36 ohm	360 ohm	3.6 k	36 k	360 k	3.6 M		
3.9 ohm	39 ohm	390 ohm	3.9 k	39 k	390 k	3.9 M		
4.3 ohm	43 ohm	430 ohm	4.3 k	43 k	430 k	4.3 M		
4.7 ohm	47 ohm	470 ohm	4.7 k	47 k	470 k	4.7 M		
5.1 ohm	51 ohm	510 ohm	5.1 k	51 k	510 k	5.1 M		
5.6 ohm	56 ohm	560 ohm	5.6 k	56 k	560 k	5.6 M		
6.2 ohm	62 ohm	620 ohm	6.2 k	62 k	620 k	6.2 M		
6.8 ohm	68 ohm	680 ohm	6.8 k	68 k	680 k	6.8 M		
7.5 ohm	75 ohm	750 ohm	7.5 k	75 k	750 k	7.5 M		
8.2 ohm	82 ohm	820 ohm	8.2 k	82 k	820 k	8.2 M		
9.1 ohm	91 ohm	910 ohm	9.1 k	91 k	910 k	9.1 M		

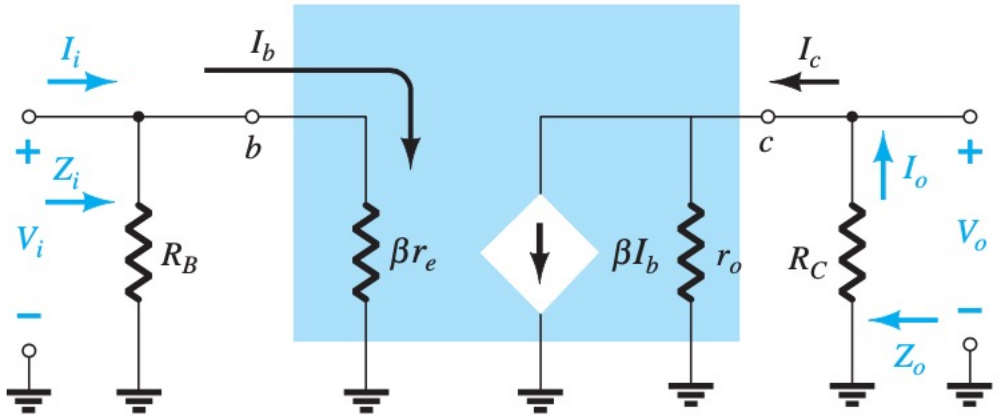
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Rangkaian Skematik Percobaan



Analisis AC Fixed Bias





$$Z_i = R_B \| \beta r_e \quad \text{ohms}$$

$$Z_i \cong \beta r_e \qquad R_B \geq 10\beta r_e \qquad \text{ohms}$$

$$V_o = -\beta I_b (R_C \| r_o)$$

$$I_b = \frac{V_i}{\beta r_e}$$

$$V_o = -\beta \left(\frac{V_i}{\beta r_e} \right) (R_C \| r_o)$$

$$A_v = \frac{V_o}{V_i} = -\frac{(R_C \| r_o)}{r_e}$$

$$A_v = -\frac{R_C}{r_e} \qquad r_o \geq 10R_C$$

Langkah dan Tugas Percobaan

1. Desainlah rangkaian BJT fixed bias dengan konfigurasi common emitter
2. Dengan memanfaatkan informasi pada datasheet, hitunglah nilai V_{CC}, R_C, R_B, dan I_B untuk kondisi daerah aktif jika diinginkan nilai $I_{CQ} \cong 5mA$ dan $V_{CEQ} \cong 5V$
3. Rangkailah rangkaian BJT fixed bias dengan konfigurasi common emitter sesuai dengan desain yang telah anda buat (manfaatkan nilai R yang ada dipasaran dengan melakukan pendekatan perhitungan)
4. Berikan input AC sebesar 10mVpp dan f = 1KHz amati sinyal input dan output menggunakan oscilloscope
5. Hitunglah penguatan yang terjadi berdasarkan pengukuran
6. Bandingkan pengutan tegangan berdasarkan perhitungan dan pengukuran; hitunglah %error yang terjadi
7. Clues($I_{CQ} = ½*I_{Csat}$; $V_{CEQ} = ½*V_{CC}$; nilai hfe/β kemungkinan ≅ nilai rata-rata dari min-max β; $r_o=h_{oe}$ dari datasheet)